Introduction to CI/CD:

CI : Continuous Integration
CD : Continuous Delivery
CD : Continuous Deployment

It is a modern software development practice that automates the process of building, testing, and releasing applications.

It plays a key role in DevOps by streaming collaboration between development and operations teams.

- Automates code integration, testing, and deployment workflows.
- Reduce manual efforts while improving software quality.
- Smaller, frequent updates replace large risky releases.

Before CI/CD adoption, software development and delivery were slow, manual, and prone to errors. Processes were less automated, making releases time-consuming and risky.

After CI/CD adoption:

- Developers commit code frequently to a shared repository.
- CI performs build and test automatically.

Three Pillars of CI/CD:

1. Continuous Integration
2. Continuous Delivery
3. Continuous Deployment

Continuous Integration:
It focuses on integrating code changes frequently to avoid conflicts and ensure code stability.
Goal : Prevent “Integration Hell” caused by late code merging.
Process : Developers merge code changes into the main branch frequently.
Automation : Each commit triggers an automated build and unit tests.
Outcome : If test fails, the build is rejected and developers are notified immediately.

Continuous Delivery:
It ensures that the application is always ready to release, with minimal manual efforts.
Goal : Keep the codebase in a release ready state at all times.
Process : After CI passes, code is deployed to a staging or testing environment.
Automation : Integration, system, and performance tests are executed automatically.
Release : Deployment to production is manual, typically triggered when needed.

Continuous Deployment:
It takes automation a step further by removing manual intervention in releases.

Goal : Enable fully automated and faster production releases.
Process : After all tests pass, code is automatically deployed to production.
Automation : End to End pipeline runs without human involvement.
Requirement : Requires highly reliable and comprehensive automated testing.

CI WorkFlow: